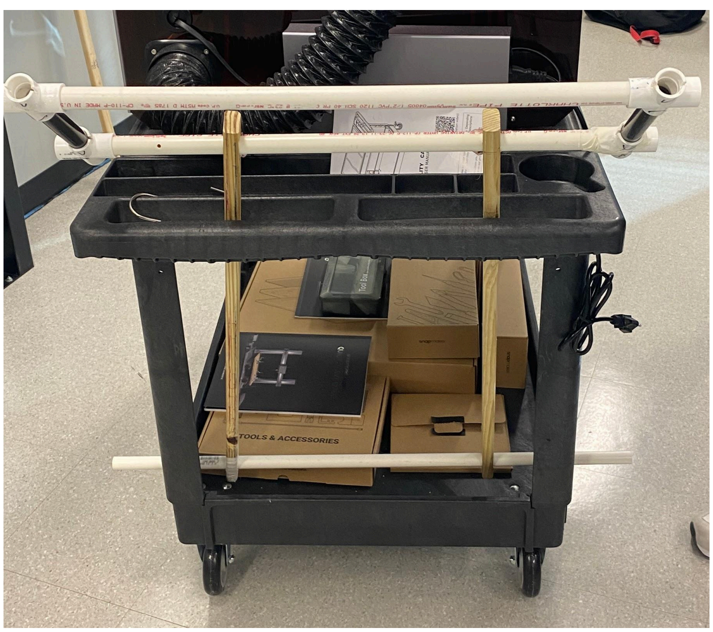
Element K

A couple weeks ago, we were given the task of creating a handle design that fit our stakeholder, Mr. Kiser, better than the original candidate. After researching past handle adjustment types, we started the initial concept design. We each came up with many designs then created a design matrix to pick the most viable option. Now that we have one main design, we started creating the base prototype out of wood rods and PVC pipes. This was where we met our first challenge. While drilling a hole into one of the wood rods, a piece of the wood broke off.

Instead of completely restarting our project, we decided to grow from this mistake and try to incorporate it into our main design. We created two ‘towers’ that would allow for the stakeholder to hold the rods however high they wanted as there was no horizontal rod, just two vertical wooden planks. Then, one day when we were testing this hook mechanism on the cart, the wood broke again showing that it wasn't strong enough to be a practical solution.



And with this, we had to continue with another prototype. So we tried to divert back to the design with holes for PVC rods to slide through at the base and for the handles as shown in the images above. Also, on one Monday, many faculty members and students from Vanderbilt came to help us continue to design and iterate our already existing prototype. Dr. London really helped us decide how to create the best base for our project, and als she gave us the idea to add instructions on the side so our clients know how to take off our prototype when we aren't there to help. A student there named Jared gave us some ideas for a mechanism that would allow our prototype to lock into place at different intervals. Though we didn't completely use this, the use of rubber to allow the handle to stay up was based on this idea.

Finally, we have the best version of our original design so far. We added more PVC pipes to increase the height reach of the handle. We put rubber onto the pipe so that there is friction at the attachment point so the handle can stay up even without anyone touching it. Through all the testing of different designs and iterations inspired by staff members and college students, we have managed to create a successful handle for our client, Mr. Kiser to use and hopefully eliminate the back pain he has experienced in the past.